

Discovering Hidden Process Views from Translucent Event Logs

Evaluation

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Abstract

We evaluate the relationship-induced abstraction κ in three steps: whether it *recovers* the hidden view-defining context, whether its per-view models are *better* than global discovery, trace clustering, and variant discovery, and how it behaves in a realistic *case study* where the recorded enabled activities carry noise. Across nine synthetic logs κ recovers the hidden context exactly (ARI 1.00), parameter-free, and its per-view models are the most faithful to the enabled behaviour among the compact approaches. Its one weakness is made explicit: κ reads structure only from the enabled sets, so corrupting them degrades it sharply while control-flow clustering is unaffected but never competitive.

1 Evaluation

We evaluate the relationship-induced abstraction κ (Section 4) in three steps that mirror the three claims of the introduction: whether κ *recovers* the hidden view-defining context, whether its per-view models are *better* than global discovery, trace clustering, and variant discovery, and how it behaves in a realistic *case study* where the recorded enabled activities carry noise. All experiments use synthetic translucent event logs, so the hidden context is known and recovery can be measured exactly; reported numbers are means over three random seeds. Our implementation and the evaluation pipeline are publicly available;¹ everything is implemented in Python with PM4Py [7].

1.1 Experimental setup

Synthetic logs. Each generator fixes one control-flow skeleton over a small activity alphabet — a spine of activities with a few *flexible blocks* (a set of choices, an optional step) — and a *hidden label* that sets how each block behaves: a block of choices is resolved concurrently, in a fixed order, or is pinned to one branch; and an optional step is present or absent. To generate a case we draw one hidden label and emit a random trace consistent with it, recording at every event both the executed activity and the set of activities that were selectable (its enabled set); the label is then removed, so it survives only in the pattern of enabled sets, which is what κ reads. Each generator is tuned so that all variants of one hidden label collapse onto a *single* κ identifier while different labels get different identifiers. Table 1 lists the nine logs; each isolates one phenomenon, stated in its group header. The *k-view* family is the controlled sweep — one base process in two to five configurations — and varies the ground-truth k ; each of the other five logs adds one axis k -view does not exercise. *Running example* is the paper’s running example L_{run} at scale: eight activities — open (o), a check (b or c), an assessment (d or e), submit (f), approve (g) or reject (h) — whose hidden label offers the two choices concurrently, sequentially, or pinned to one branch. Scaling its six traces to 1500 cases leaves the three abstraction identifiers unchanged, so κ must return exactly $\text{seq}(o, \text{and}(\text{xor}(b, c), \text{xor}(d, e)), f, \text{xor}(g, h))$, $\text{seq}(o, \text{xor}(b, c), \text{xor}(d, e), f, \text{xor}(g, h))$, and $\text{seq}(o, b, d, f, \text{xor}(g, h))$. *Parallel choices* keeps the executed sequences identical across labels, so only the enabled sets can separate them and control-flow clustering cannot succeed even in principle; *enterprise* scales the setting to ~ 20 activities and five roles; *role split* is the minimal clean case (six classical variants) and the only log where a hidden label skips activities entirely; and *rework loop* has no hidden context at all — the honest failure, where κ over-segments an unbounded loop. The column **hidden** k is the number of distinct hidden labels the generator used (“–” when there is no hidden context) and κ **views** is the number of equivalence classes κ induces; the two coincide on every log with a hidden context and diverge only for *rework loop*, which has none and where κ unrolls the loop into four length-indexed views.

Baselines. We compare κ with eight alternatives. *Global* discovery mines one model from the whole log with the translucent Inductive Miner (IMtf) [11]. *Variant discovery* groups traces by their executed projection [4, 5], and *translucent-variant grouping* refines this to one group per distinct translucent trace. *Control-flow trace clustering* (CF-TC) [1] clusters activity-count and directly-follows features with k -means and Ward linkage;

¹<https://github.com/hherbertb/translucent-process-views>

Table 1: The nine synthetic logs and the phenomenon each isolates (group headers). Right of the rule is a result: κ produces exactly hidden- k views on every log except the rework loop.

log	cases	acts	avg len	class. var.	transl. var.	hidden k	κ views
<i>running example – L_{run} from the preliminaries at scale; the hidden label offers the two choices concurrently, sequentially, or pinned</i>							
running example	1500	8	5.0	16	26	3	3
<i>k-view – one base process, k compositions of three choices; does κ track the number of views without being told k?</i>							
k -view ($k=2$)	800	8	5.0	48	56	2	2
k -view ($k=3$)	1200	8	5.0	48	57	3	3
k -view ($k=4$)	1600	8	4.2	49	58	4	4
k -view ($k=5$)	1800	9	4.6	57	66	5	5
<i>parallel choices – labels differ only in which pair is concurrent; executed sequences are identical</i>							
parallel choices	2000	13	8.0	390	535	5	5
<i>enterprise – ~ 20 activities, five roles toggling three review blocks; holds at realistic size</i>							
enterprise	3000	21	11.8	2304	2503	5	5
<i>role split – three roles, only six classical variants; the smallest clean case and the only log where a label skips activities entirely</i>							
role split	1040	7	5.6	6	8	3	3
<i>rework loop – no hidden context; κ has no loop operator and unrolls it (the honest failure)</i>							
rework loop	600	5	5.0	4	4	–	4

context-aware clustering (CTX-TC) [2] adds enabled-activity features and *enabled-only* clustering (EN-TC) uses those alone. *Representation-learning* clustering (D2V-TC) [3] embeds each trace with Doc2Vec [14], and *attribute* clustering (ATTR-TC) clusters on recorded case attributes: four *clean* ones that carry information about the hidden label — **role** has normalised mutual information 1.00 with it, **department** 0.55, **location** 0.47, **seniority** 0.19 — and six *noisy* ones independent of it (all ≤ 0.003). The tables list the k -means and Ward variants of CF-TC and the clean / noisy / all variants of ATTR-TC separately, each run with k set to the ground-truth number of views (GT) and with k chosen by the silhouette [13] (auto).

Metrics. Each partition is compared with the hidden-label partition by the adjusted Rand index (ARI), normalised mutual information (NMI), and purity; all three lie in $[0, 1]$ and are 1 for a perfect match. *Enabled-signature incoherence* is the mean number of distinct per-position enabled-set sequences per (group, classical variant) pair; it is 1 iff no group mixes availability contexts. For *model quality* we discover one model per group with IMtf and score it on its own sub-log: fitness (alignments [8]), precision (alignment-based [6]), generalisation, simplicity (net size), and the enabled-set-aware translucent fitness / precision [9, 10]; F1 is the harmonic mean of fitness and precision. Because several models are discovered per log, the per-log score is the case-frequency-weighted mean over the groups and the reported value is the unweighted mean over the nine logs (the case-weighted variant is in Table 5). The model-quality experiment uses one seed; the partition metrics average three.

1.2 Recovering hidden context

Table 2 reports, for every grouping and given the ground-truth k , whether its groups are the right ones. Global discovery yields one group and recovers nothing; variant and translucent-variant grouping produce 325 and 368 groups for a hidden k of at most five, and their groups still mix 1.24 enabled-activity signatures per (group, variant) against 1.00 for κ . Given $k = \text{GT}$, the control-flow, context-aware, enabled-only, and Doc2Vec baselines reach $\text{ARI} \leq 0.53$ (and ≤ 0.65 with k by silhouette), whereas κ reaches ARI, NMI, and purity of 1.00 with no cluster count, distance function, threshold, or clustering algorithm. On the running example this is the paper’s own answer: the three identifiers κ induces are exactly the three listed above, one per hidden resolution, at ARI 1.00 and 500 cases each. The only baseline that matches κ is attribute clustering on the clean attributes (ARI 1.00) — and only because **role** was recorded and k was supplied; on the noisy attributes it is at chance.

1.3 Model quality

Table 3 compares the per-group models on all eight metrics. Fitness is 1.00 for every approach, because each model is discovered from and scored on the same sub-log, so the classic axes reduce to precision and F1. Among the approaches that use neither a recorded attribute nor k , κ is the most precise (precision 0.97, F1 0.98) at 3.8 models; variant discovery matches it on classic precision (0.98) only by discovering one trivial model per variant — about 325 of them — which gives it the best simplicity (0.99) and the worst generalisation (0.77,

Table 2: Recovering the hidden context: ARI / NMI / purity against the hidden label, and enabled-signature incoherence, for every grouping. *GT* / *auto* = *k* set to the true view count / chosen by silhouette. Mean over the nine logs and three seeds; the best value in each column is in **bold**.

	ARI	NMI	purity	enabl. sig.
global	0.000	0.000	0.303	1.236
κ (ours)	1.000	1.000	1.000	1.000
CF-TC <i>k</i> -means (GT)	0.238	0.323	0.542	1.265
CF-TC <i>k</i> -means (auto)	0.297	0.398	0.644	1.236
CF-TC Ward (GT)	0.331	0.446	0.636	1.265
CF-TC Ward (auto)	0.340	0.476	0.700	1.236
CTX-TC (GT)	0.351	0.508	0.645	1.246
CTX-TC (auto)	0.371	0.552	0.632	1.173
EN-TC (GT)	0.514	0.651	0.733	1.125
EN-TC (auto)	0.644	0.733	0.850	1.034
D2V-TC (GT)	0.470	0.556	0.689	1.061
D2V-TC (auto)	0.324	0.467	0.563	1.110
ATTR-TC clean (GT)	1.000	0.999	1.000	1.000
ATTR-TC clean (auto)	0.984	0.983	0.996	1.000
ATTR-TC noisy (GT)	0.176	0.218	0.501	1.066
ATTR-TC noisy (auto)	0.239	0.305	0.607	1.037
ATTR-TC all (GT)	0.863	0.912	0.921	1.009
ATTR-TC all (auto)	0.760	0.856	0.809	1.015
variant discovery	0.251	0.400	0.852	1.236
translucent-variant grouping	0.311	0.522	1.000	1.000

versus ≈ 0.94 for κ and the clustering baselines). The enabled-set-aware measures separate the approaches: κ reaches translucent fitness 0.98 and translucent F1 0.99, the best of all approaches, whereas the single global model admits enabled behaviour that no hidden view exhibits (0.89 / 0.94) and variant discovery scores 0.86 / 0.92 despite its trivial classic precision. κ ’s per-view models are thus the most faithful to the enabled behaviour, at an order of magnitude fewer models than variant discovery.

1.4 Case study: sensitivity to noise

Setup. We model an IT service-desk desktop application. The base workflow is `open` $\rightarrow$ a *triage* block `{categorize, set_priority}` $\rightarrow$ a *diagnosis* block `{check_kb, run_diag, remote}` $\rightarrow$ an *action* `xor(apply_fix, request_info)` with a gated `escalate` or `approve` $\rightarrow$ a *close* `xor(resolve, close_noaction, force_close)` $\rightarrow$ `log_time, notify`. The hidden context is the operator’s role: an *agent* runs triage sequentially, uses one diagnosis step, and can only resolve; a *senior* runs triage and a three-way diagnosis in parallel and may escalate; a *specialist* skips triage and close entirely; a *supervisor* runs triage in parallel and has `approve` and `force_close` on screen. The enabled activities are read off screenshots [12], so they carry noise: with probability p , one enabled activity per event is added or dropped (the executed activity is never touched). We generate 1000 cases, balanced over the four roles, at $p \in \{0, 0.05, 0.1, 0.15, 0.2\}$ with three seeds.

Figure 1(a) is the de-jure model: the four roles share one skeleton and differ only in how each block is constrained — an agent runs triage in sequence and can only `resolve`, a senior and a supervisor run triage in parallel, a specialist drops triage and close, and only a supervisor sees `approve` and `force_close`. The rest of Figure 1 contrasts one global model with the four models κ induces at $p = 0$. The global model is visibly loose — it makes triage optional, turns diagnosis into a three-way parallel block with skips, and merges the action and close steps into one seven-way choice with a loop back — whereas each of κ ’s models matches one role exactly: the agent model is a plain sequence, the senior model runs triage and a wide diagnosis block in parallel and offers `escalate`, the specialist model has neither a triage nor a close step, and only the supervisor model has `approve` and `force_close`.

Table 4 shows the effect of noise. At $p = 0$, κ recovers all four roles exactly (ARI 1.00, four views); as p rises its ARI falls to 0.31 and 0.08 and its view count grows from 4 to 412 and 659, because each corrupted trace becomes its own view. The control-flow baselines behave oppositely: their partition is read from the executed activities, which noise never touches, so their ARI stays around 0.6–0.85 across all noise levels — and never recovers the roles, which differ mainly in availability, not execution. Attribute clustering on the clean attributes stays at ARI 1.00 throughout, since `role` is unaffected by enabled-set noise — again, only because it was recorded and k was supplied. κ ’s translucent F1 nevertheless stays high (1.00 then ≈ 0.94) as its ARI collapses: each noise fragment is tiny and still modelled faithfully; what degrades is the grouping, not the per-group models.

Limitations. The synthetic logs give κ a clean hidden context and thus its best case; real contexts overlap and drift. κ depends entirely on the fidelity of the enabled sets, as the case study shows, and reaches parity

Table 3: Model quality per approach, mean over the nine logs. *#mdl* = mean number of models; *fit.* = fitness (always 1.00); *gen.* = generalisation; *simp.* = simplicity; *tr.* = translucent. The best value in each column is in **bold** (*#mdl* and *fit.* excepted).

	#mdl	fit.	prec.	F1	gen.	simp.	tr. fit.	tr. prec.	tr. F1
global	1.000	1.000	0.938	0.964	0.962	0.696	0.891	0.989	0.936
κ (ours)	3.778	1.000	0.969	0.984	0.939	0.815	0.978	0.997	0.987
CF-TC <i>k</i> -means (GT)	3.778	1.000	0.954	0.976	0.939	0.789	0.907	0.996	0.949
CF-TC <i>k</i> -means (auto)	5.222	1.000	0.924	0.959	0.928	0.820	0.901	0.999	0.947
CF-TC Ward (GT)	3.778	1.000	0.922	0.959	0.937	0.785	0.926	0.995	0.959
CF-TC Ward (auto)	5.222	1.000	0.947	0.972	0.932	0.838	0.927	0.999	0.962
CTX-TC (GT)	3.778	1.000	0.904	0.937	0.936	0.793	0.897	0.961	0.927
CTX-TC (auto)	4.222	1.000	0.914	0.944	0.936	0.812	0.913	0.959	0.934
EN-TC (GT)	3.778	1.000	0.860	0.911	0.941	0.801	0.902	0.918	0.908
EN-TC (auto)	5.556	1.000	0.918	0.948	0.929	0.826	0.927	0.961	0.944
D2V-TC (GT)	3.778	1.000	0.827	0.899	0.935	0.752	0.888	0.920	0.903
D2V-TC (auto)	2.889	1.000	0.856	0.915	0.946	0.753	0.880	0.932	0.904
ATTR-TC clean (GT)	3.750	1.000	0.972	0.985	0.932	0.792	0.985	1.000	0.992
ATTR-TC clean (auto)	4.625	1.000	0.980	0.990	0.937	0.799	0.992	0.999	0.996
ATTR-TC noisy (GT)	3.750	1.000	0.865	0.926	0.931	0.698	0.903	0.979	0.939
ATTR-TC noisy (auto)	6.750	1.000	0.750	0.854	0.896	0.709	0.876	0.940	0.907
ATTR-TC all (GT)	3.750	1.000	0.970	0.984	0.938	0.766	0.972	1.000	0.985
ATTR-TC all (auto)	3.250	1.000	0.932	0.963	0.941	0.742	0.940	0.975	0.957
variant discovery	324.667	1.000	0.983	0.991	0.768	0.994	0.861	0.997	0.924
translucent-variant grouping	368.111	1.000	0.983	0.991	0.738	0.994	0.861	0.997	0.924

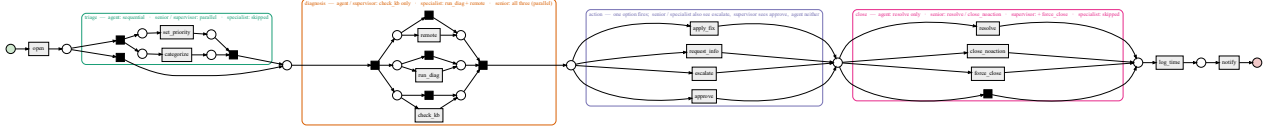
Table 4: Service-desk case study: ARI and translucent F1 for every approach at each enabled-set noise level p (mean over three seeds); the best value in each column is in **bold**.

	ARI $p=0$	tr. F1 $p=0$	ARI $p=0.05$	tr. F1 $p=0.05$	ARI $p=0.1$	tr. F1 $p=0.1$	ARI $p=0.15$	tr. F1 $p=0.15$	ARI $p=0.2$	tr. F1 $p=0.2$
global	0.000	0.566	0.000	0.715	0.000	0.661	0.000	0.624	0.000	0.652
κ (ours)	1.000	1.000	0.573	0.935	0.311	0.947	0.154	0.939	0.075	0.938
CF-TC <i>k</i> -means (GT)	0.629	0.985	0.699	0.948	0.738	0.945	0.707	0.935	0.720	0.909
CF-TC <i>k</i> -means (auto)	0.460	0.874	0.362	0.880	0.434	0.918	0.430	0.938	0.497	0.925
CF-TC Ward (GT)	0.640	0.985	0.775	0.963	0.746	0.973	0.745	0.933	0.736	0.894
CF-TC Ward (auto)	0.460	0.874	0.405	0.867	0.548	0.956	0.536	0.955	0.550	0.949
CTX-TC (GT)	0.630	0.985	0.849	0.968	0.837	0.977	0.814	0.946	0.774	0.944
CTX-TC (auto)	0.587	0.931	0.497	0.907	0.333	0.853	0.333	0.819	0.333	0.823
EN-TC (GT)	0.629	0.985	0.584	0.912	0.484	0.918	0.581	0.863	0.707	0.890
EN-TC (auto)	0.714	0.988	0.608	0.920	0.385	0.873	0.544	0.869	0.221	0.773
D2V-TC (GT)	0.735	0.910	0.409	0.894	0.561	0.814	0.592	0.855	0.617	0.873
D2V-TC (auto)	0.452	0.854	0.498	0.917	0.491	0.886	0.504	0.810	0.490	0.791
ATTR-TC clean (GT)	1.000	1.000	1.000	0.977	1.000	0.926	1.000	0.962	1.000	0.952
ATTR-TC clean (auto)	0.997	0.999	0.937	0.958	0.914	0.912	0.906	0.899	0.888	0.942
ATTR-TC noisy (GT)	0.114	0.638	0.085	0.671	0.033	0.715	0.025	0.693	0.002	0.717
ATTR-TC noisy (auto)	0.229	0.746	0.072	0.615	0.055	0.686	0.021	0.672	0.007	0.682
ATTR-TC all (GT)	0.704	0.877	1.000	0.977	1.000	0.926	0.999	0.943	0.999	0.944
ATTR-TC all (auto)	0.642	0.729	1.000	0.977	0.957	0.914	0.999	0.943	0.982	0.940
variant discovery	0.224	0.949	0.226	0.920	0.227	0.931	0.228	0.922	0.229	0.934
translucent-variant grouping	0.271	0.949	0.141	0.946	0.071	0.942	0.035	0.938	0.015	0.934

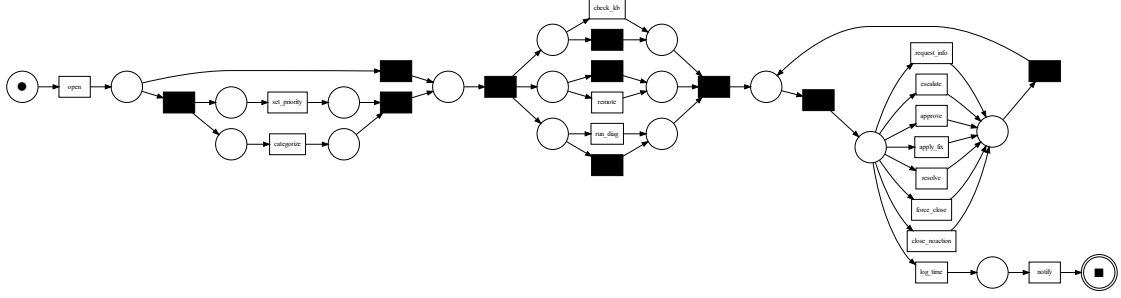
with attribute slicing once the context is recorded and k is known. It has no loop operator and over-segments unbounded rework (the *rework loop* log: four views for zero hidden contexts), all models use a single miner (IMtf), and a few very loose sub-models fall back to token-based replay and earn no translucent-quality credit. Overall, κ recovers hidden availability context exactly and parameter-free when the enabled sets are trustworthy, and its per-view models are the most faithful to the enabled behaviour; its reliance on that single signal is also its main weakness.

A Additional detail

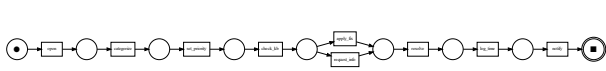
Figure 2 resolves the case study by metric — κ 's adjusted Rand index collapses as noise rises while its translucent fitness and F1 stay high, and the control-flow baselines are flat throughout. Table 5 is the case-weighted variant of Table 3: weighting the nine logs by their case counts does not change the ordering.



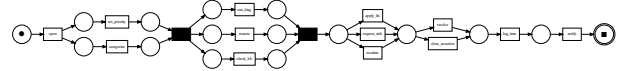
(a) The designed (de-jure) workflow — the union of what all four roles can do — annotated with how each role restricts the triage, diagnosis, action, and close blocks.



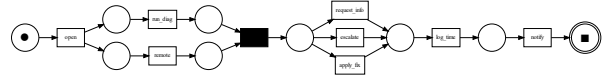
(b) One model mined from the whole log (roles conflated).



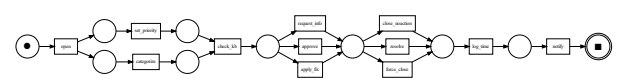
(c) κ view — agent.



(d) κ view — senior.



(e) κ view — specialist.



(f) κ view — supervisor.

Figure 1: Service-desk workflow at $p = 0$: (a) the designed (de-jure) model, the union of all four roles; (b) one model mined over the whole log; (c–f) the four models κ induces, one per hidden role. Places are circles, transitions are boxes, silent transitions are filled.

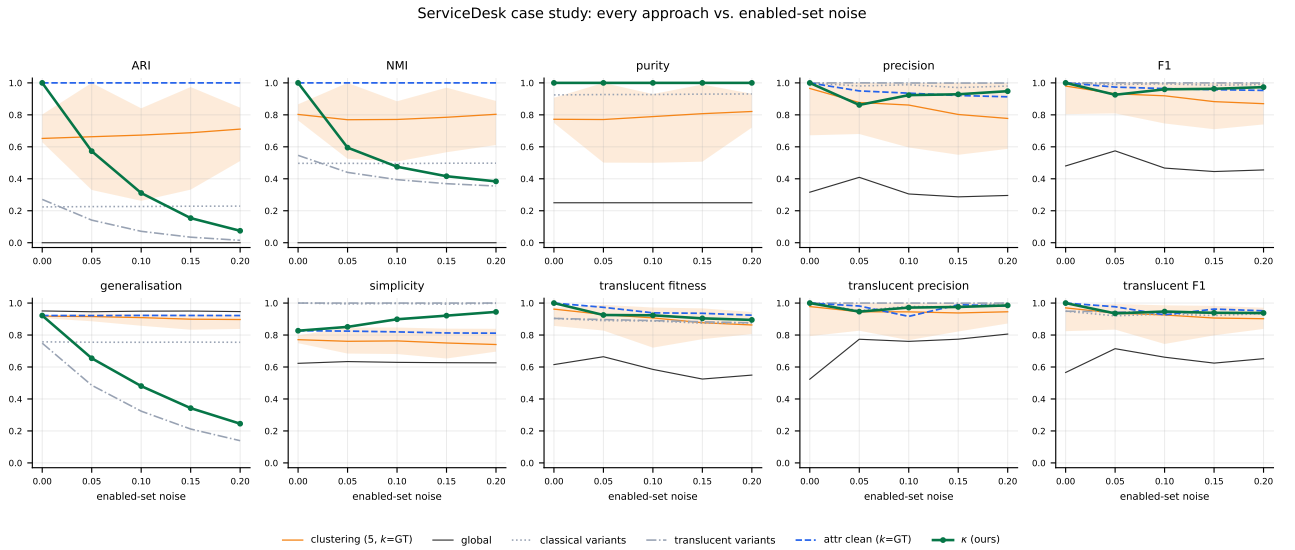


Figure 2: Service-desk case study: each metric against the enabled-set noise level p . κ (bold); the five $k = \text{GT}$ clustering baselines as a min–max band with their mean; *ATTR-TC clean* ($k = \text{GT}$), *variant discovery*, and *global* as named lines.

Table 5: Case-weighted variant of Table 3: each log weighted by its case count.

	#mdl	fit.	prec.	F1	gen.	simp.	tr. fit.	tr. prec.	tr. F1
global	1.000	1.000	0.944	0.969	0.966	0.679	0.888	0.983	0.932
κ (ours)	4.108	1.000	0.967	0.982	0.940	0.786	0.987	0.999	0.992
CF-TC k -means (GT)	4.108	1.000	0.951	0.974	0.940	0.760	0.913	0.998	0.953
CF-TC k -means (auto)	4.786	1.000	0.936	0.966	0.935	0.785	0.912	0.999	0.953
CF-TC Ward (GT)	4.108	1.000	0.917	0.956	0.939	0.757	0.932	0.994	0.962
CF-TC Ward (auto)	4.786	1.000	0.951	0.974	0.939	0.800	0.937	1.000	0.967
CTX-TC (GT)	4.108	1.000	0.881	0.920	0.936	0.761	0.895	0.951	0.922
CTX-TC (auto)	4.157	1.000	0.895	0.931	0.940	0.777	0.911	0.946	0.927
EN-TC (GT)	4.108	1.000	0.863	0.911	0.942	0.773	0.910	0.923	0.915
EN-TC (auto)	5.448	1.000	0.901	0.937	0.934	0.789	0.928	0.949	0.938
D2V-TC (GT)	4.108	1.000	0.805	0.884	0.934	0.723	0.884	0.908	0.895
D2V-TC (auto)	2.806	1.000	0.845	0.906	0.950	0.723	0.875	0.923	0.897
ATTR-TC clean (GT)	4.113	1.000	0.960	0.979	0.933	0.772	0.985	1.000	0.992
ATTR-TC clean (auto)	5.334	1.000	0.965	0.982	0.937	0.778	0.992	0.999	0.995
ATTR-TC noisy (GT)	4.113	1.000	0.843	0.913	0.932	0.683	0.901	0.971	0.934
ATTR-TC noisy (auto)	6.852	1.000	0.750	0.855	0.902	0.690	0.882	0.943	0.911
ATTR-TC all (GT)	4.113	1.000	0.958	0.978	0.939	0.748	0.973	1.000	0.986
ATTR-TC all (auto)	3.578	1.000	0.915	0.954	0.942	0.720	0.941	0.973	0.956
variant discovery	590.963	1.000	0.993	0.996	0.678	0.998	0.852	0.999	0.919
translucent-variant grouping	661.264	1.000	0.993	0.996	0.642	0.998	0.852	0.999	0.919

References

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